

Linear Equation with Constant Coefficients.

The general form of linear differential equation with Constant coefficients of order n is,

$$\frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_1 \frac{dy}{dx} + a_0 y = Q(x) \quad \text{or } f(x)$$

or,

$$y^n + a_{n-1} y^{n-1} + a_{n-2} y^{n-2} + \dots + a_1 y + a_0 y = Q(x) \quad \text{or } f(x) \quad \text{--- (1)}$$

where, $a_0, a_1, a_2, \dots, a_{n-1}$ are constant and Q is constant or function of x .

Equation (1) is non-homogenous.

If $Q(x) = 0$, then equation (1) becomes homogenous.

$$y^n + a_{n-1} y^{n-1} + a_{n-2} y^{n-2} + \dots + a_1 y + a_0 y = 0 \quad \text{--- (2)}$$

The general solution of (1) is given by,

$$\boxed{y = y_{C.F} + y_{P.I}} \Rightarrow y = C.F + P.I$$

The general solution of (2) is given by,

$$\boxed{y = y_{C.F}} \quad \text{Here, } y_{P.I} = 0$$

Homogenous Linear Differential Equation with Constant Coeff.

The homogenous linear differential with constant coefficient is of the form,

$$\frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_1 \frac{dy}{dx} + a_0 y = 0 \quad \rightarrow (1)$$

OR

$$(D^n + a_{n-1} D^{n-1} + a_{n-2} D^{n-2} + \dots + a_1 D + a_0) y = 0 \text{ where } D^n = \frac{d^n y}{dx^n}$$

where $a_0, a_1, a_2, a_3, \dots, a_{n-1}$ are coefficient.

Equation of first order: Simplest case $\rightarrow$

Let $n=1$, we obtain,

$$a_1 \frac{dy}{dx} + a_0 y = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{a_0}{a_1} y$$

$$\Rightarrow \frac{dy}{y} = -\frac{a_0}{a_1} dx$$

$$\Rightarrow \ln|y| = -\frac{a_0}{a_1} x + \ln|c|$$

$$\Rightarrow y = c e^{-\frac{a_0}{a_1} x}$$

$$\Rightarrow \boxed{y = A e^{mx}}$$

so, we shall call $y = A e^{mx}$ a trial solution ~~and~~ of the equation

(1).

Now, suppose that $y = A e^{mx}$ be the trial solⁿ of (1), then

$$A m^n e^{mx} + A a_{n-1} m^{n-1} e^{mx} + A a_{n-2} m^{n-2} e^{mx} + \dots + A a_0 e^{mx} = 0$$

$$\Rightarrow A e^{mx} (m^n + a_{n-1} m^{n-1} + a_{n-2} m^{n-2} + \dots + a_1 m + a_0) = 0$$

Since, $e^{mx} \neq 0$

so,

$$m^n + a_{n-1}m^{n-1} + a_{n-2}m^{n-2} + \dots + a_1m + a_0 = 0 \rightarrow (2).$$

Equation (2) is known as auxiliary equation (A.E).

How to find Complementary function (C.F)

Rule-01.

Let $m_1, m_2, \dots, m_n$ be n distinct roots of A.E, then

C.F is, $y_{C.F} = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$

where $c_1, c_2, \dots, c_n$ are arbitrary constants.

Rule-02

If two roots m_1 & m_2 of the A.E are equal, each equal to m and $m_3, m_4, \dots, m_n$ are distinct then C.F is,

$$y = (c_1 + c_2 x) e^{mx} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

If three roots m_1, m_2 & m_3 of the A.E are equal, each equal to m and $m_4, m_5, \dots, m_n$ are distinct then C.F is,

$$y = (c_1 + c_2 x + c_3 x^2) e^{mx} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

and so on.

If all roots are equal say m , then C.F is,

$$y = (c_1 + c_2 x + c_3 x^2 + \dots + c_n x^{n-1}) e^{mx}$$

Rule-03.

If the A.E has a pair of imaginary roots, say $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$ and $m_3, m_4, \dots, m_n$ are distinct then C.F is,

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$\text{or, } y = c_1 e^{\alpha x} \cos(\beta x + c_2) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$\text{or } y = c_1 e^{\alpha x} \sin(\beta x + c_2) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Any one of these forms may be used.

Rule-04.

If a pair of imaginary roots $\alpha \pm i\beta$ occurs twice and $m_5, m_6, \dots, m_n$ are distinct then C.F is,

$$y = e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x] + c_5 e^{m_5 x} + \dots + c_n e^{m_n x}$$

Rule-05.

If the two real roots be m and $-m$ and $m_3, m_4, \dots, m_n$ are distinct then C.F is,

$$y = (c_1 e^{mx} + c_2 e^{-mx}) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$\text{or, } = (c_1 \cosh mx + c_2 \sinh mx) + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

$$\text{or, } = [c_1 (\cosh mx + \sinh mx) + c_2 (\cosh mx - \sinh mx)] + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

Problem

$$(1) \text{ Solve: } 2 \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + y = 0 \longrightarrow (1)$$

Soln $\rightarrow$ Let $y = e^{mx}$ be the trial soln of (1), then A.E is,

$$2m^2 - 3m + 1 = 0$$

$$\Rightarrow 2m^2 - (2+1)m + 1 = 0$$

$$\Rightarrow (2m-1)(m-1) = 0$$

$$\Rightarrow m = 1, \frac{1}{2}$$